

# Introduction

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# Disclaimer

Discussions taken from Alonso Sepúlveda [1], Landau & Lifschits [2], Lautrup [3], Lai-Rubin-Kreml [4] books

## 1 Preliminaries

Fluid mechanics may be seen as a starting point for field theory, both from the mathematical and physical perspectives. This is not only because, historically, the study of fluids created the language with which fields are described, but also because it is one of the first courses in a physics major where field theory can be applied. Hydrodynamics, on the one hand, introduces fundamental concepts such as field, flux, source, sink, divergence, curl, circulation, and vorticity. On the other hand, Euler's dynamical formulation provides the framework to derive and describe results such as Torricelli's and Bernoulli's theorems, as well as the theorems on vortex circulation.

Fluid mechanics is encountered not only in almost every area of our physical lives. Blood flows through our veins and arteries, a ship moves through water, airplanes fly in the air, air flows around wind machines, air is compressed in a compressor, steam flows around turbine blades, a dam holds back water, air is heated and cooled in our homes, and computers require air to cool components.

## 2 Continuous matter

Matter is formed of molecules, which in turn consist of atoms and subatomic particles. Thus, matter is not continuous. However, beyond length scales greater than  $10^{-8}$  m matter can be described with theories that pay no attention to the molecular structure of materials (see Fig. 1). The theory that aims to describe relationships among gross phenomena, neglecting the structure of material on a smaller scale, is known as continuum theory. The continuum theory regards matter as indefinitely divisible. Thus, within this theory, one accepts the idea of an infinitesimal volume of materials, referred to as a particle in the continuum, and in every neighborhood of a particle there are always neighboring particles.

- It is usually assumed that the physical laws valid at one length scale are not very sensitive to the details of what happens at much smaller scales.
- The microscopic physics affects the macroscopic world almost only through material constants, such as coefficients of elasticity and viscosity, characterizing the interactions between macroscopic amounts of matter. It is, of course, an important task for the physics of materials to derive the values of these constants, but this task lies outside the realm of continuum physics.
- Here we consider the transition from molecules to continuous matter, or mathematically speaking from point particles to fields.
- It is emphasized that the macroscopic continuum description must necessarily be statistical in nature, but that random statistical fluctuations are strongly suppressed by the enormity of the number of molecules in any macroscopic material object.

## DIFFERENT SCALING STRUCTURE OF MATTER

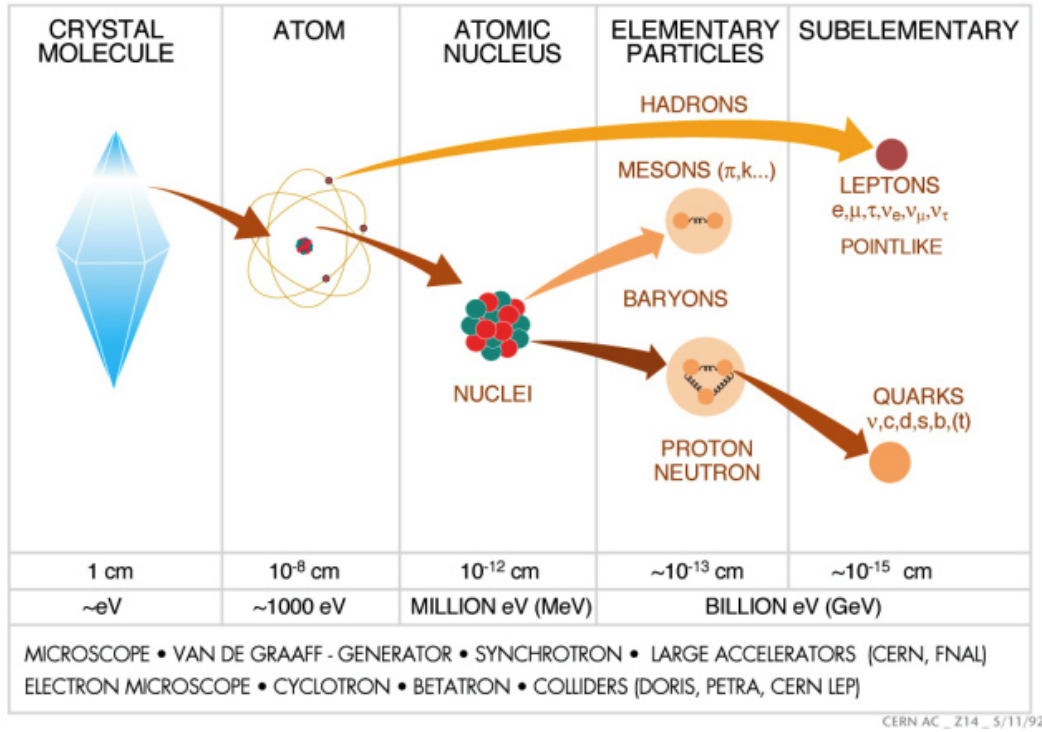


Figure 1: Different scales in the structure of matter. Figure taken from Ref. [5].

- We will recast Newton's laws for point particles into a systematic theory of continuous matter, and the application of this theory to the macroscopic material world.

### 2.1 Mole and molar mass

As of May 20, 2019, with the redefinition of SI base units, Avogadro's number is defined as an *exact constant*:

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}.$$

This means that one mole corresponds to exactly

$$6.02214076 \times 10^{23}$$

specified elementary entities. Unlike the pre-2019 definition, it is no longer based on the mass of 12 g of carbon-12, but rather fixed in the same way that the speed of light  $c$  is defined in SI units.

The *molar mass* of a substance is defined as the mass of one mole of its particles (atoms, molecules, or ions). By definition, one mole corresponds to  $N_A$  particles. The molar mass is expressed in *grams per mole* ( $\text{g/mol}$ ). The molar mass  $M$  can be expressed as

$$M_{\text{mol}} = \frac{m}{n}, \quad (1)$$

where  $m$  is the mass of the sample (in grams) and  $n$  is the amount of substance (in moles).

- The molar mass of hydrogen (H) is approximately

$$M(\text{H}) \simeq 1.008 \text{ g/mol}.$$

- The molar mass of water ( $\text{H}_2\text{O}$ ) is given by

$$M(\text{H}_2\text{O}) = 2 \times 1.008 + 15.999 = 18.015 \text{ g/mol}.$$

In short, the molar mass specifies the mass of  $6.022 \times 10^{23}$  entities of a substance.

### 2.1.1 The Kilogram

Before May 20, 2019, the kilogram was defined as the mass of the *International Prototype of the Kilogram (IPK)*, a cylinder of platinum–iridium kept at the International Bureau of Weights and Measures (BIPM) in Sèvres, France.

The kilogram is now defined in terms of the *Planck constant*,  $h$ , which has been assigned the exact value

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s} = \text{kg} \cdot \text{m}^2/\text{s}. \quad (2)$$

From this fixed constant, and using the definitions of the second and the meter, the kilogram is derived. In practice, the realization of the kilogram is achieved through experiments such as the *Kibble balance* or the *x-ray crystal density method*.

## 2.2 Molecular separation length

Let a sample of a pure substance with volume  $V$  and mass  $m$ . The molar mass is denoted as  $m = n \times M_{\text{mol}}$ , where  $n = N/N_A$  with  $N$  being the number of molecules. The volume per molecule is  $V_{\text{mol}} = V/N$ , and assuming that  $V_{\text{mol}} = L_{\text{mol}}^3$  we have that the molecular separation length becomes

$$L_{\text{mol}} = \left( \frac{V}{N} \right)^{1/3} = \left( \frac{M_{\text{mol}}}{\rho N_A} \right)^{1/3}. \quad (3)$$

This length defines the scale at which the discrete molecular nature of matter governs the physics, rendering any continuum description of bulk matter entirely invalid.

For liquids and solids where the molecules touch each other, this length is roughly the size of a molecule. For solid iron  $L_{\text{mol}} \approx 0.23 \text{ nm}$ , and for liquid water  $L_{\text{mol}} \approx 0.31 \text{ nm}$ . For an ideal gas at normal temperature and pressure ( $p = 1 \text{ atm} = 101325 \text{ Pa}$  and  $T = 20^\circ\text{C} = 293.15 \text{ K}$ ) and with

$$pV = nRT = n(N_A k_B)T = N k_B T, \quad (4)$$

we obtain

$$L_{\text{mol}} = \left( \frac{k_B T}{p} \right)^{1/3} = \left( \frac{1.380649 \times 10^{-23} \text{ J/K } 293.15 \text{ K}}{101325 \text{ N m}^{-2}} \right)^{1/3} \quad (5)$$

$$\approx 3.42 \text{ nm}. \quad (6)$$

There is a lot of vacuum in a volume of gas, in fact about 1000 times the true volume of the molecules at normal temperature and pressure.

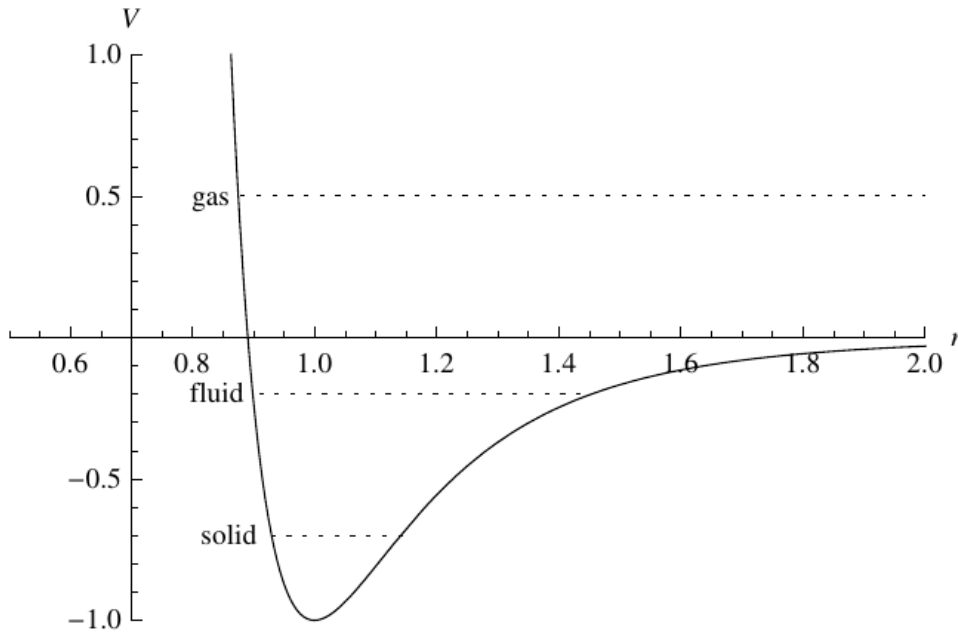


Figure 2: Sketch of the intermolecular potential energy  $V(r)$  between two roughly spherical neutral molecules as a function of the distance  $r$  between their centers. It is attractive at moderate range and strongly repulsive at close distance. The equilibrium between attraction and repulsion is found at the minimum of the potential. Here the units are arbitrarily chosen such that the equilibrium distance becomes  $r = 1$  and the minimum potential  $V(1) = -1$ . The horizontal dashed lines suggest the thermal energy levels for solid, liquid, and gaseous matter. Figure taken from [3].

## 2.3 States of matter

Material interactions are fundamentally electromagnetic, appearing in the form of van der Waals forces between neutral atoms and molecules. In general, these forces are strongly repulsive when molecules are pushed closer than their natural size allows, while they become moderately attractive when molecules are separated. The competition between short-range repulsion and long-range attraction establishes an equilibrium distance comparable to the molecular size. This balance of forces is commonly modeled by the well-known Lennard–Jones potential (see Fig. 2)

$$V(r) = 4\varepsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right], \quad (7)$$

where  $\varepsilon$  characterizes the depth of the potential well,  $\sigma$  sets the equilibrium distance, and  $r$  is the interparticle separation.

Particles may arrange themselves in a number of different ways to minimize their total energy. The total energy receives negative contributions from the intermolecular potential energy, which attempts to bind the molecules to each other near equilibrium, and positive contributions from the kinetic energy in their thermal motion. The three classic states of neutral matter—solid, liquid, and gas—depend, broadly speaking, on the competition between negative binding energy and positive thermal energy.

### 2.3.1 Solids

*In solid matter the binding is so strong that thermal motion cannot overcome it. The molecules remain bound to each other by largely elastic forces, and constantly undergo small-amplitude thermal motion around their equilibrium positions. If increasing external forces are applied, solids will begin to deform elastically, until they eventually become plastic or even fracture. A solid body retains its shape independently of the shape of a container large enough to hold it, apart from small deformations, for example due to gravity.*

### 2.3.2 Liquids

*In liquid matter the binding is weaker than in solid matter, although it is still hard for a molecule on its own to leave the company of the others through an open liquid surface. The molecules stay in contact but are not locked to their neighbors. Molecular conglomerates may form and stay loosely connected for a while, as for example chains of water molecules. Under the influence of external forces, for example gravity, a liquid will undergo bulk motion, called flow, a process that may be viewed as a kind of continual fracturing. A liquid will not expand to fill an empty container completely, but will under the influence of external forces eventually adapt to its shape wherever it touches it.*

### 2.3.3 Gases

*In gaseous matter the molecules are bound so weakly that the thermal motion easily overcomes it, and they essentially move around freely between collisions. A gas will always expand to fill a closed empty container completely. Under the influence of external forces, for example a piston pushed into the container, a gas will quickly flow to adapt to the changed container shape.*

### 2.3.4 Plasmas

*A plasma is an ionized gas consisting of free electrons and ions. Because of the presence of charged particles, plasmas exhibit collective electromagnetic behavior and can sustain electric currents, magnetic fields, and waves. They constitute more than 99% of the visible universe, being found in stars, nebulae, and the interstellar medium. On Earth, plasmas appear in lightning, flames, and fusion devices.*

## 2.4 Granular matter

Granular matter (like grains, sand, or building materials) consists of discrete elements whose behavior is strongly influenced by friction. Unlike fluids, heaps of grains do not spread out but can form stable structures such as sand ripples or sandcastles. While this course does not treat granular matter explicitly, at sufficiently large scales it may approximate continuous matter, regardless of whether the grains are microscopic or cosmic in size.

As an example, on scales exceeding approximately 100 Mpc, the large-scale distribution of galaxies is well approximated by a continuous fluid description, as illustrated in Fig. 3.

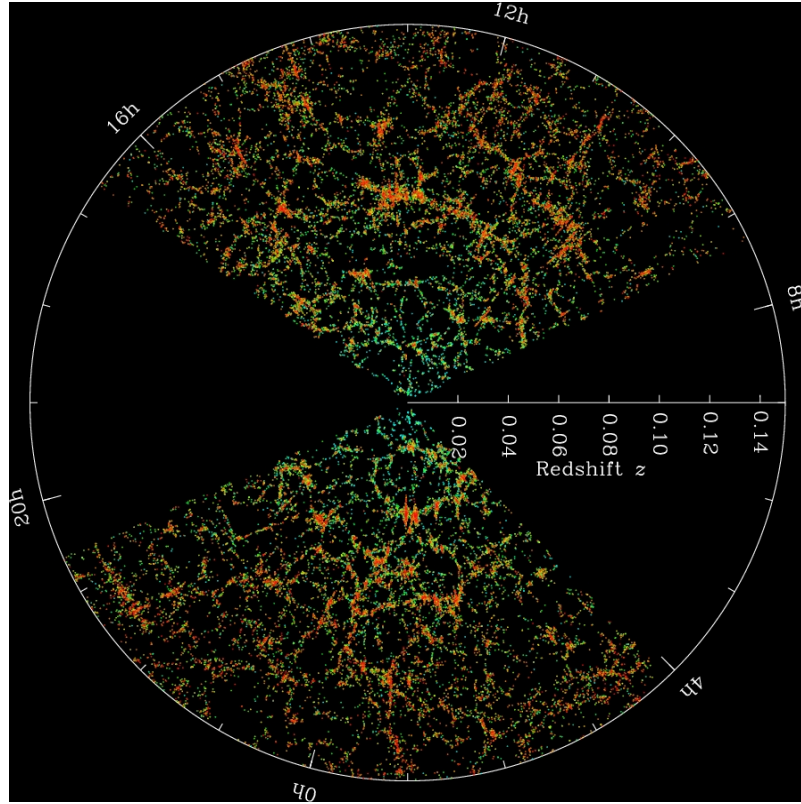


Figure 3: A map of galaxy positions in a narrow slice of the Universe, obtained by the Sloan Digital Sky Survey. The Milky Way is at the center, and the survey extends to a radius of about 600 Mpc. Galaxy locations are determined from the redshift of their spectral lines. Credit: M. Blanton and the Sloan Digital Sky Survey.

## 3 The continuum approximation

### 3.1 Poisson distribution

The *Poisson distribution* models the probability of observing a certain number of events  $n$  in a fixed interval of time, space, or volume under the following conditions:

- Events occur independently of one another.
- The average rate of occurrence is constant.
- The probability of more than one event in an infinitesimal interval is negligible.

The probability mass function is

$$P(n|\lambda) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad n = 0, 1, 2, \dots$$

where  $\lambda$  is the expected (mean) number of events in the interval. This distribution is commonly used to describe “rare events,” such as the number of radioactive decays in a sample during a given time, the number of molecules found in a small subvolume of a gas, or the number of phone calls received in a fixed time window.

Let's consider a small volume  $V$  of a much larger volume of gas, such that the probability for any molecule to be found in  $V$  is exceedingly small. If the average number of molecules in  $V$  is known to be  $N$ , the probability of finding precisely  $n$  of the molecules in  $V$  is given

$$\Pr(n|N) = \frac{N^n}{n!} e^{-N}, \quad n = 0, 1, 2, \dots$$

It is normalized:

$$\sum_{n=0}^{\infty} \Pr(n|N) = e^{-N} \sum_{n=0}^{\infty} \frac{N^n}{n!} = e^{-N} e^N = 1.$$

The mean value is given by

$$\langle n \rangle = \sum_{n=0}^{\infty} n \Pr(n|N) = e^{-N} \sum_{n=1}^{\infty} \frac{n N^n}{n!} = e^{-N} \sum_{n=1}^{\infty} \frac{N^n}{(n-1)!}.$$

Let  $m = n - 1$ :

$$\langle n \rangle = e^{-N} \sum_{m=0}^{\infty} \frac{N^{m+1}}{m!} = e^{-N} N \sum_{m=0}^{\infty} \frac{N^m}{m!} = e^{-N} N e^N = N.$$

To calculate the variance it is convenient to compute  $\langle n(n-1) \rangle$ :

$$\langle n(n-1) \rangle = e^{-N} \sum_{n=2}^{\infty} \frac{n(n-1) N^n}{n!} = e^{-N} \sum_{n=2}^{\infty} \frac{N^n}{(n-2)!}.$$

Let  $m = n - 2$ :

$$\langle n(n-1) \rangle = e^{-N} \sum_{m=0}^{\infty} \frac{N^{m+2}}{m!} = e^{-N} N^2 \sum_{m=0}^{\infty} \frac{N^m}{m!} = N^2.$$

Now, since  $\langle n^2 \rangle = \langle n(n-1) \rangle + \langle n \rangle$ , we have

$$\langle n^2 \rangle = N^2 + N.$$

Therefore the variance is

$$\Delta N^2 = \text{Var}(n) = \langle n^2 \rangle - \langle n \rangle^2 = (N^2 + N) - N^2 = N,$$

so that  $\Delta N = \sqrt{N}$ .

### 3.2 Density fluctuations

Whether a given number of molecules is large enough to warrant the use of a smooth continuum description of matter depends on the desired precision. Since matter is never continuous at sufficiently high precision, continuum physics is always an approximation. But as long as the fluctuations in physical quantities caused by the discreteness of matter are smaller than the desired precision, matter may be taken to be continuous. To observe the continuity, one must so to speak avoid looking too sharply at material bodies.

Consider a fixed small volume  $V$  of a pure gas with molecules of mass  $m$ . If at a given time  $t$  the number of molecules in this volume is  $N$ , the mass density at this time becomes

$$\rho(t) = \frac{N m}{V}. \quad (8)$$



At a time  $t' = t + \Delta t$  (with  $\Delta t$  much larger than the time interval  $\tau$  between molecular collisions):

$$\rho(t') = \frac{N' m}{V}. \quad (9)$$

Since the density is linear in  $N$ , the relative fluctuation in density becomes

$$\frac{\Delta \rho}{\rho} = \frac{\Delta N}{N} = \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}. \quad (10)$$

For  $N \sim N_A$ , the relative fluctuation becomes of the order of  $10^{-12}$  and can safely be ignored.

If we want the relative density fluctuation to be smaller than a given value  $\Delta \rho / \rho \lesssim \epsilon$ , we must require  $N \gtrsim \epsilon^{-2}$ . The smallest acceptable number of molecules,  $\epsilon^{-2}$ , occupies a volume  $\epsilon^{-2} L_{\text{mol}}^3$ . A cubic cell with this volume has side length

$$L_{\text{micro}} = \epsilon^{-2/3} L_{\text{mol}}. \quad (11)$$

Thus, to secure a relative precision  $\epsilon = 10^{-3}$  for the density, the microscopic cell should contain  $10^6$  molecules and have side length  $L_{\text{micro}} = 100 L_{\text{mol}}$ . For an ideal gas under normal conditions we find  $L_{\text{micro}} \approx 0.34 \mu\text{m}$ .

## Macroscopic smoothness

As a criterion for a smooth continuum description we demand that the relative variation in density between neighboring cells should be less than the desired precision  $\epsilon$ . The change in density between the centers of neighboring cells along some direction  $x$  can be estimated by Taylor expanding  $\rho(x')$ :

$$\rho(x') = \rho(x + \Delta x) = \rho(x) + \frac{d\rho}{dx} \Delta x + \dots \quad (12)$$

Identifying  $\Delta x = L_{\text{micro}}$  we obtain

$$\Delta \rho \approx L_{\text{micro}} \left| \frac{\partial \rho}{\partial x} \right|. \quad (13)$$

Demanding the relative variation to be smaller than the precision,  $\Delta \rho / \rho \lesssim \epsilon$ , we obtain a constraint on the magnitude of the density derivative,

$$\left| \frac{\partial \rho}{\partial x} \right| \lesssim \frac{\rho}{L_{\text{macro}}}, \quad (14)$$

where

$$L_{\text{macro}} = \epsilon^{-1} L_{\text{micro}}. \quad (15)$$

As long as the above condition is fulfilled, the density may be considered to vary smoothly, because the density changes over the micro-scale are imperceptible. Any significant change in density must take place over distances larger than  $L_{\text{macro}}$ . With  $\epsilon = 10^{-3}$  we find  $L_{\text{macro}} \approx 1000 L_{\text{micro}} \approx 0.34 \text{ mm}$  for an ideal gas under normal conditions.

The thickness of interfaces between macroscopic bodies is typically on the order of  $L_{\text{mol}}$  and thus much smaller than  $L_{\text{macro}}$ . Consequently, these regions of space fall outside the smooth continuum description. In continuum physics, interfaces appear instead as surface discontinuities in the otherwise smooth macroscopic description of matter.

### 3.3 Velocity

#### 3.3.1 Maxwell–Boltzmann speed distribution $f(v)$

Let's consider a ideal, classical and nonrelativistic gas. For one molecule of mass  $m$ , the kinetic energy is  $E = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2)$ . In thermal equilibrium at temperature  $T$ , the canonical (Boltzmann) weight is  $\propto e^{-E/(k_B T)}$ . By independence and isotropy,

$$p(\mathbf{v}) d^3v = A \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z, \quad (16)$$

with  $A$  fixed by normalization  $\int_{\mathbb{R}^3} p(\mathbf{v}) d^3v = 1$ . Each component is Gaussian with variance  $\sigma^2 = k_B T/m$ :

$$\int_{-\infty}^{\infty} e^{-mv_x^2/(2k_B T)} dv_x = \sqrt{\frac{2\pi k_B T}{m}}, \quad (17)$$

and similarly for  $v_y, v_z$ . Hence

$$1 = \int_{\mathbb{R}^3} p(\mathbf{v}) d^3v = A \left( \sqrt{\frac{2\pi k_B T}{m}} \right)^3 \implies A = \left( \frac{m}{2\pi k_B T} \right)^{3/2}. \quad (18)$$

Thus the *velocity-vector* density is

$$p(\mathbf{v}) = \left( \frac{m}{2\pi k_B T} \right)^{3/2} \exp\left(-\frac{mv^2}{2k_B T}\right), \quad v = \|\mathbf{v}\|. \quad (19)$$

The probability that the speed lies in  $[v, v + dv]$  equals the probability that  $\mathbf{v}$  lies in the spherical shell of radius  $v$  and thickness  $dv$ :  $d^3v = 4\pi v^2 dv$ . Define the *speed* density  $f(v)$  by  $f(v) dv = \int_{\text{shell}} p(\mathbf{v}) d^3v$ . Using (19),

$$f(v) = 4\pi v^2 \left( \frac{m}{2\pi k_B T} \right)^{3/2} \exp\left(-\frac{mv^2}{2k_B T}\right), \quad v \geq 0. \quad (20)$$

#### 3.3.2 Speeds from the Maxwell–Boltzmann Distribution

The MB speed distribution for a 3D ideal gas is

$$f(v) = 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right), \quad v \geq 0, \quad (21)$$

normalized so that  $\int_0^\infty f(v) dv = 1$ .

- Most probable speed  $v_{\text{mp}}$ .

Maximize  $f(v)$  with respect to  $v$ :

$$\frac{d}{dv}[\ln f(v)] = \frac{d}{dv}\left[2 \ln v - \frac{mv^2}{2k_B T}\right] = 0 \implies \frac{2}{v} - \frac{m}{k_B T}v = 0, \quad (22)$$

$$\implies v_{\text{mp}}^2 = \frac{2k_B T}{m}, \quad \boxed{v_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}}. \quad (23)$$

- Mean speed  $\langle v \rangle$ .

Compute  $\langle v \rangle = \int_0^\infty v f(v) dv$ . Using  $\int_0^\infty v^n e^{-\alpha v^2} dv = \frac{1}{2} \alpha^{-(n+1)/2} \Gamma\left(\frac{n+1}{2}\right)$ , with  $\alpha = \frac{m}{2k_B T}$ :

$$\langle v \rangle = 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 \exp(-\alpha v^2) dv \quad (24)$$

$$= 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \cdot \frac{1}{2} \alpha^{-2} \Gamma(2) = 2\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \alpha^{-2}. \quad (25)$$

Since  $\alpha^{-2} = (2k_B T/m)^2$  and  $\Gamma(2) = 1$ ,

$$\langle v \rangle = 2\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \left( \frac{2k_B T}{m} \right)^2 = \sqrt{\frac{8k_B T}{\pi m}}, \quad \boxed{\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}}. \quad (26)$$

- Root-mean-square speed  $v_{\text{rms}}$ .

Compute  $\langle v^2 \rangle = \int_0^\infty v^2 f(v) dv$ :

$$\langle v^2 \rangle = 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4 \exp(-\alpha v^2) dv \quad (27)$$

$$= 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \cdot \frac{1}{2} \alpha^{-5/2} \Gamma\left(\frac{5}{2}\right) = 2\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \alpha^{-5/2} \cdot \frac{3\sqrt{\pi}}{4}. \quad (28)$$

With  $\alpha = \frac{m}{2k_B T}$ , this simplifies to

$$\langle v^2 \rangle = \frac{3k_B T}{m}, \quad v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \boxed{\sqrt{\frac{3k_B T}{m}}}. \quad (29)$$

Using  $m = M_{\text{mol}}/N_A$  and  $R_{\text{mol}} = N_A k_B$ :

$$v_{\text{mp}} = \sqrt{\frac{2R_{\text{mol}} T}{M_{\text{mol}}}}, \quad \langle v \rangle = \sqrt{\frac{8R_{\text{mol}} T}{\pi M_{\text{mol}}}}, \quad v_{\text{rms}} = \sqrt{\frac{3R_{\text{mol}} T}{M_{\text{mol}}}}. \quad (30)$$

They satisfy the universal ordering

$$v_{\text{mp}} < \langle v \rangle < v_{\text{rms}}, \quad (31)$$

with fixed ratios (independent of  $T$  and the gas species):

$$\frac{v_{\text{mp}}}{v_{\text{rms}}} = \sqrt{\frac{2}{3}} \approx 0.8165, \quad \frac{\langle v \rangle}{v_{\text{rms}}} = \sqrt{\frac{8}{3\pi}} \approx 0.9213, \quad \frac{\langle v \rangle}{v_{\text{mp}}} = \sqrt{\frac{4}{\pi}} \approx 1.1284. \quad (32)$$

For air at room temperature,  $T = 300$  K and  $M_{\text{mol}} \approx 0.029$  kg mol<sup>-1</sup>. Then

$$v_{\text{mp}} = \sqrt{\frac{2R_{\text{mol}} T}{M_{\text{mol}}}} = \sqrt{\frac{2 \times 8.314 \times 300}{0.029}} \approx 4.14 \times 10^2 \text{ m s}^{-1}, \quad (33)$$

$$\langle v \rangle = \sqrt{\frac{8R_{\text{mol}} T}{\pi M_{\text{mol}}}} = \sqrt{\frac{8 \times 8.314 \times 300}{\pi \times 0.029}} \approx 4.68 \times 10^2 \text{ m s}^{-1}, \quad (34)$$

$$v_{\text{rms}} = \sqrt{\frac{3R_{\text{mol}} T}{M_{\text{mol}}}} = \sqrt{\frac{3 \times 8.314 \times 300}{0.029}} \approx 5.07 \times 10^2 \text{ m s}^{-1}. \quad (35)$$

### 3.3.3 Derivation of the root-mean-square average of the center-of-mass velocity

Consider a collection of  $N$  identical molecules (a “material particle”) taken from a large volume of gas. Let the instantaneous molecular velocities be  $\mathbf{v}_n$  for  $n = 1, 2, \dots, N$ . Collisions with other molecules in the gas at large will randomly change the velocity of each of the selected molecules, but if there is no overall drift in the gas, the velocity of individual molecules should average out to zero,  $\langle \mathbf{v}_n \rangle = \mathbf{0}$ ; the velocities of different molecules should be uncorrelated,  $\langle \mathbf{v}_n \cdot \mathbf{v}_m \rangle = 0$  for  $n \neq m$ ; and the average of the square of the velocity should be the same for all molecules,  $\langle v_n^2 \rangle = v_0^2$ . The center-of-mass velocity is

$$\mathbf{V}_{\text{cm}} = \frac{1}{N} \sum_{n=1}^N \mathbf{v}_n. \quad (36)$$

Its mean-square value is

$$\langle V_{\text{cm}}^2 \rangle = \left\langle \left( \frac{1}{N} \sum_{n=1}^N \mathbf{v}_n \right) \cdot \left( \frac{1}{N} \sum_{m=1}^N \mathbf{v}_m \right) \right\rangle = \frac{1}{N^2} \sum_{n,m=1}^N \langle \mathbf{v}_n \cdot \mathbf{v}_m \rangle. \quad (37)$$

Using the assumptions, the cross terms vanish for  $n \neq m$  and  $\langle \mathbf{v}_n \cdot \mathbf{v}_n \rangle = \langle v_n^2 \rangle = v_0^2$ , so

$$\langle V_{\text{cm}}^2 \rangle = \frac{1}{N^2} \sum_{n=1}^N v_0^2 = \frac{N v_0^2}{N^2} = \frac{v_0^2}{N}. \quad (38)$$

Therefore the root-mean-square center-of-mass speed is

$$V_{\text{rms}} = \sqrt{\langle V_{\text{cm}}^2 \rangle} = \frac{v_0}{\sqrt{N}} \quad (39)$$

$$\equiv \Delta v. \quad (40)$$

### 3.3.4 Velocity fluctuations

Since the fluctuation scale is set by the molecular velocity,  $v_{\text{mol}} = v_{\text{rms}}$ , it takes much larger numbers of molecules to be able to ignore the fluctuations in everyday gas velocities. To maintain a relative precision  $\epsilon$  in the velocity fluctuations we must require  $\Delta v/v \lesssim \epsilon$ , implying that the linear size of a gas volume must be larger than

$$L'_{\text{micro}} = \left( \frac{v_{\text{mol}}}{v} \right)^{2/3} L_{\text{micro}}.$$

The velocity fluctuations of a gentle steady wind, say  $v \approx 0.5 \text{ m s}^{-1}$ , can be ignored with precision  $\epsilon \approx 10^{-3}$  for volumes of linear size larger than  $L'_{\text{micro}} = 100 L_{\text{micro}} \approx 34 \text{ } \mu\text{m}$ . The smoothness scale should similarly be  $L'_{\text{macro}} = \epsilon^{-1} L'_{\text{micro}}$ , which in this case becomes  $L'_{\text{macro}} \approx 34 \text{ mm}$ . A hurricane wind,  $v \approx 50 \text{ m s}^{-1}$ , only requires volumes of linear size  $L'_{\text{micro}} = 4.6 L_{\text{micro}} \approx 1.6 \text{ } \mu\text{m}$  to yield the desired precision, but in this case fluctuations due to turbulence will anyway completely swamp the molecular fluctuations.

## Mean free path

Another condition for obtaining a valid continuum description is that molecules should interact with each other to “iron out” strong differences in velocities. If there were no interactions, a molecule with a given velocity would keep on moving with that velocity forever. In solids and liquids where the molecules are closely packed, these interactions take place over a molecular separation length.

In a gas every molecule traces out a straight path through the vacuum until it collides with another molecule. The *mean free path*  $\lambda$  is defined as the average distance traveled by a single molecule. The other molecule is, however, not a “sitting duck” but travels itself on average also a distance  $\lambda$  before it is hit by the first. The mean free path turns out to be

$$\lambda = \frac{L_{\text{mol}}^3}{\sqrt{2} \pi d^2} = \frac{M_{\text{mol}}}{\sqrt{2} \pi d^2 \rho N_A} = 0.225 \frac{m}{d^2 \rho}, \quad (41)$$

where  $d$  is the molecular diameter and  $m$  is the mass of a molecule. Since  $\lambda/L_{\text{mol}} \sim \rho^{-2/3}$ , the mean free path will in sufficiently dilute gases always become larger than the micro scale and should instead be used to define the smallest linear scale for the continuum description.

Air consists mainly of nitrogen and argon molecules with an average molar mass  $M_{\text{mol}} \approx 29 \text{ g mol}^{-1}$  and average diameter  $d \approx 0.37 \text{ nm}$ . At normal temperature and pressure the mean free path becomes  $\lambda \approx 65 \text{ nm}$ , which is five times smaller than the microscopic length scale for the density,  $L_{\text{micro}} \approx 340 \text{ nm}$  (for  $\epsilon = 10^{-3}$ ). The mean collision time may be estimated as  $\tau \approx \lambda/v_{\text{mol}}$ , and becomes for air  $\tau \approx 0.13 \text{ ns}$ .

For air  $m = 4.8 \times 10^{-26} \text{ kg}$  and  $d = 3.7 \times 10^{-10} \text{ m}$ . At standard atmospheric conditions the mean free path is approximately  $6.4 \times 10^{-6} \text{ cm}$ , at an elevation of 100 km it is 10 cm, and at 160 km it is 5000 cm. Obviously, at higher elevations the continuum assumption is not acceptable and the theory of rarified gas dynamics (or free molecular flow) must be utilized.

Calculate the mean free path in the atmosphere at an elevation of:

- (a) 30 000 m
- (b) 50 000 m
- (c) 80 000 m

## 4 Fields

In a mathematical sense, material particles are taken to be truly infinitesimal and all physical properties of the particles as well as other physical observables are described by smooth functions of space and time. Continuum physics is therefore a theory of fields.

Mathematically, a field  $\phi$  is simply a real-valued function.

$$\phi = \phi(\vec{x}, t) = \phi(x_1, x_2, x_3, t) = \phi(x, y, z, t). \quad (42)$$

In general, The fields can be classified as scalar, vector or tensor.

## 4.1 Scalar field

In this case, the field associates a specific number to every single point:

$$\phi = \phi(\vec{x}, t) = \phi(x_1, x_2, x_3, t) = \phi(x, y, z, t). \quad (43)$$

As examples of scalar fields we have:

- Density field:  $\rho(\vec{x}, t)$ .
- Temperature field:  $T(\vec{x}, t)$ .

The generalization to special relativity is

$$\phi = \phi(\vec{x}, ct) = \phi(x_1, x_2, x_3, x_4) = \phi(x^\mu). \quad (44)$$

## 4.2 Vector fields

In this case, the field associates a specific vector to every single point:

$$\vec{E} = \vec{E}(\vec{x}, t) = \vec{E}(x_1, x_2, x_3, t) = \vec{E}(x, y, z, t). \quad (45)$$

As examples of vector fields we have:

- Gravitational, electric and magnetic fields:  $\vec{g}(\vec{x}, t)$ ,  $\vec{E}(\vec{x}, t)$  and  $\vec{B}(\vec{x}, t)$ .
- Velocity field:  $\vec{v}(\vec{x}, t)$ .

The generalization to special relativity is

$$E^\nu = E^\nu(x^\mu). \quad (46)$$

## 4.3 Tensor fields

In this case, the field associates a specific tensor to every single point:

$$\sigma_{ij} = \sigma_{ij}(\vec{x}, t) = \sigma_{ij}(x_1, x_2, x_3, t) = \sigma_{ij}(x, y, z, t). \quad (47)$$

As examples of tensor fields we have:

- Stress tensor:  $\sigma_{ij}(\vec{x}, t)$ .

The generalization to special relativity is

$$F^{\alpha\beta} = F^{\alpha\beta}(x^\mu). \quad (48)$$

## 4.4 Field equations

The dynamics of physical fields are governed by *field equations*, typically formulated as partial differential equations in both space and time. In continuum mechanics, the fundamental equations of motion arise from the local balance laws of mass, momentum, and energy, obtained by applying Newton's Second Law and the First Law of Thermodynamics to each material particle within the continuum. The resulting system comprises the continuity equation, the Navier–Stokes equations for momentum, and an energy balance, supplemented by constitutive relations for stresses and heat fluxes.

For a Newtonian fluid of density  $\rho(\mathbf{x}, t)$ , velocity field  $\mathbf{v}(\mathbf{x}, t)$ , and specific internal energy  $e(\mathbf{x}, t)$ , the governing equations are:

- Continuity (mass conservation):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (49)$$

- Navier–Stokes (momentum balance):

$$\rho \left( \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v} + \left( \zeta + \frac{1}{3} \mu \right) \nabla (\nabla \cdot \mathbf{v}) + \rho \mathbf{f}, \quad (50)$$

- Energy (first law of thermodynamics):

$$\rho \left( \frac{\partial e}{\partial t} + \mathbf{v} \cdot \nabla e \right) = -p \nabla \cdot \mathbf{v} + \Phi + \nabla \cdot (k \nabla T) + \rho q, \quad (51)$$

where:

- $p$  is the pressure,
- $\mu$  the shear viscosity,
- $\zeta$  the bulk viscosity,
- $\mathbf{f}$  external body forces (e.g. gravity),
- $\Phi$  the viscous dissipation function,
- $k$  the thermal conductivity,
- $T$  the absolute temperature,
- $q$  a possible volumetric heat source.

In contrast to linear field theories such as Maxwell's equations of electromagnetism, Eqs. (49)–(51) are intrinsically nonlinear. The principal source of this nonlinearity is the advective term in the momentum and energy equations, e.g.

$$(\mathbf{v} \cdot \nabla) \mathbf{v},$$

which represents the self-interaction of the velocity field. This quadratic coupling makes the velocity field simultaneously the transported quantity and the agent of transport. As a consequence, the mathematical structure of continuum mechanics is profoundly more complex than that of linear systems.

Historically, the inviscid limit was first formulated by Euler in the mid-18th century, leading to what are now called the Euler equations of fluid motion. In the early 19th century, Navier and Stokes independently extended this framework by incorporating viscous stresses, arriving at the modern Navier–Stokes equations. This refinement established the foundation of fluid mechanics as a field theory and revealed the essential role of viscosity in damping and dissipating motion.

The nonlinear character of the Navier–Stokes equations underlies a range of rich dynamical phenomena, including the emergence of instabilities, bifurcations, and the transition to turbulence. Turbulent flows, in particular, are characterized by strongly nonlinear interactions across a hierarchy of scales and remain one of the central unsolved problems in classical physics. The mathematical challenge of establishing the global existence and smoothness of solutions to the three-dimensional Navier–Stokes equations is recognized as one of the seven Millennium Prize Problems of the Clay Mathematics Institute.

## 4.5 What is continuum mechanics

Although often used interchangeably, the terms have slightly different emphases:

- *Continuum mechanics*: A formal branch of mechanics that models matter as a continuous medium, ignoring its atomic structure. Physical quantities such as density, velocity, stress, and strain are treated as continuous functions of space and time. It provides a theoretical framework that includes elasticity, fluid mechanics, plasticity, and other fields.
- *Continuous mechanics*: A less common expression, often used synonymously with continuum mechanics. In physics-oriented contexts it may emphasize the study of systems made of continuous matter (e.g., fluids, solids, plasmas), rather than the abstract mathematical framework.

## 5 Problems

1. Show that a cube containing  $N = M^3$  smaller cubes of equal size will have

$$K = 6M^2 - 12M + 8$$

smaller cubes lying on the surface. Estimate the fluctuation  $\Delta N$  when  $N$  molecules in a cube oscillate with amplitude equal to the molecular size.

2. A spherical volume contains a large number  $N$  of molecules. Estimate the number of molecules  $N_S$  situated at the surface and show that the fluctuation in this number is

$$\Delta N \approx 6^{1/3} \pi^{1/6} N^{1/3} \approx 2.2 N^{1/3}$$

when they randomly oscillate with amplitude equal to the molecular size.



3. Consider a material gas particle containing  $N$  identical molecules. Write the velocity of the  $n$ th molecule as

$$\vec{v}_n = \vec{v} + \vec{u}_n$$

where  $\vec{v}$  is the centre of mass velocity and  $\vec{u}_n$  is a random contribution from thermal motion.

It may be assumed that the average of the random component of velocity vanishes  $\langle \vec{u}_n \rangle = 0$ , that all random velocities are uncorrelated, and that their fluctuations are the same for all particles  $\langle u_n^2 \rangle = v_0^2$ .

Show that the average of the centre of mass velocity for the fluid particle is

$$\langle \vec{v}_c \rangle = \vec{v}$$

and that its fluctuation due to thermal motion is

$$\Delta v_c = \frac{v_0}{\sqrt{N}}.$$

4. At what precision is the microscopic scale  $L_{\text{micro}}$  equal to the mean free path, when the air density is 100 times smaller than normal?
5. Using Eq. (3) for the molecular separation length,

$$L_{\text{mol}} = \left( \frac{M_{\text{mol}}}{\rho N_A} \right)^{1/3},$$

compute  $L_{\text{mol}}$  for:

- a) Liquid water at room temperature ( $\rho = 1000 \text{ kg/m}^3$ ).
- b) An ideal gas at  $T = 293 \text{ K}$  and  $p = 1 \text{ atm}$ .

Compare the results and discuss why the continuum approximation is more appropriate for liquids than for gases.

6. Consider a cubic microscopic cell of side length  $L_{\text{micro}}$  inside a gas at normal conditions.
- a) Derive the condition on  $N$  (number of molecules) such that the relative density fluctuation  $\Delta\rho/\rho$  is less than  $10^{-3}$ .
- b) Estimate  $L_{\text{micro}}$  for air at room temperature.
7. For nitrogen gas ( $M_{\text{mol}} = 0.028 \text{ kg/mol}$ ) at  $T = 300 \text{ K}$ , calculate:
- a) The most probable speed  $v_{\text{mp}}$ .
- b) The mean speed  $\langle v \rangle$ .
- c) The root-mean-square speed  $v_{\text{rms}}$ .

Verify the inequality  $v_{\text{mp}} < \langle v \rangle < v_{\text{rms}}$ .

8. Starting from the discrete Lagrangian for a 1D chain of masses  $m$  and springs of constant  $k$ ,

$$L = \sum_n \left[ \frac{m}{2} \dot{u}_n^2 - \frac{k}{2} (u_{n+1} - u_n)^2 \right],$$

derive in the continuum limit the 1D wave equation

$$\rho \partial_t^2 u - \tau \partial_x^2 u = 0,$$

identifying the physical meaning of  $\rho$  and  $\tau$ .

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